

Disease data newsroom: make the numbers fair

Teacher guide and worked answers

Create an accurate disease-data infographic using a checked source, comparable population measures and a cautious conclusion.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsl/inputs/LAUNCH KS4 - 2026-27.xlsx | LAUNCH Weekly - Spring | C35

Research disease statistics; create an infographic.

Directly develops source research and an infographic from C 35. The offline source card supports source/date/denominator checking; constructed local data support original chart design. No official exam or epidemiological analysis claimed.

Prior learning

Communicable and non-communicable diseases differ.

A fraction compares a part with a whole.

Graphs need labels and units.

Success evidence

Record source, publication date, data year and what a figure measures.

Calculate cases per 1000 over the same period.

Create a chart with an accurate headline, source and limitation.

Equipment

Printed shared evidence and one selected response route; pen, ruler and calculator.

Projector or offline HTML/PowerPoint. Optional word processor or spreadsheet.

Preparation

Print the shared evidence page, one chosen response route and exit page. Routes are alternatives, not a workload ladder.

Read the teacher answers before teaching. All numerical sample data in this lesson are constructed for learning.

Optional ICT: open a blank document/slide or spreadsheet. Columns A-C can contain town, cases and population; D2 formula =B2/C2*1000. Paper infographic works fully offline.

Preview the WHO source card sensitively: it concerns global deaths, with no personal discussion required. If unsuitable for a pupil, use the constructed town source card for the same source/measure audit.

Practical care

Paper and screen tasks; no biological sampling, cultures, medicine handling or personal health disclosures.

Ready to teach alternative

All evidence is included. Point, annotate, dictate or discuss quietly instead of writing or using a device.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	A headline can be numerically true and misleading
2	Retrieval	2-5	Read what the number means
3	I do	5-9	Research the source, not just the number
4	I do	9-12	Work one fair population comparison
5	I do	12-15	A count answers one useful question
6	We do	15-18	Newsroom calculation desk
7	We do	18-20	Switch the measure, test the headline
8	We do	20-22	Editors reject an unsupported cause
9	Hinge	22-25	Which caption is accurate?
10	You do	25-26	Make a useful infographic
11	You do	26-36	Your one-page newsroom brief
12	Review	36-38	An editor checks the denominator
13	Review	38-39	Publish only what the evidence supports
14	Exit	39-40	Keep the denominator
15	Exit	Within stage	Next useful step

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening A headline can be numerically true and misleading

Display the raw-count claim. Let pupils predict; an answer about proportions requires denominators. The mission is an honest infographic, not identifying a real dangerous town.

2 Retrieval Read what the number means

R1: relevant whole/population. R2: category labels, numerical axis with units, title, period and source. R3: no. Accept three key labels before adding the full source checklist.

3 I do Research the source, not just the number

Read the source card. Model: "I record WHO, the page title and URL, publication date, data year 2021, global scope and the death measure. I do not relabel 75% as a share of all people." The page was checked on 8 September 2026; infer no current trend.

4 I do Work one fair population comparison

Model $12 \div 2000 = 0.006$, then $0.006 \times 1000 = 6$. Keep new cases per 1000 people over 7 days in the final unit. This is a cumulative incidence proportion, commonly called a rate in school work; it is not cases per person-year.

5 I do A count answers one useful question

Think aloud: “C recorded 12 more cases than B. That matters for workload, but C has six times B’s population. I cannot infer the higher proportion from the count.” Point to the zero baseline.

6 We do Newsroom calculation desk

A: $24 \div 3000 \times 1000 = 8$; B = 18; C = 5. Ask pairs or a TA to verify one calculation. A spreadsheet is optional; the teacher guide supplies the exact arithmetic.

7 We do Switch the measure, test the headline

C has the greatest count (30); B has the greatest value per 1000 (18). Use the controlled count/rate toggle or draw three bars on paper. Both claims can be true; neither proves a cause.

8 We do Editors reject an unsupported cause

Expected: B had the highest recorded new cases per 1000 during the period; the cause of the difference is not established. Suggest comparable testing and exposure information as additional research; pupils do not collect biological samples.

Reveal: B had the highest recorded cases per 1000 during these 7 days. The table does not establish why.

9 Hinge Which caption is accurate?

B is correct. A confuses count and proportion. C confuses the denominator. D confuses publication and data dates.

A. C has the largest proportion because it has 30 cases. - C has 6000 people: $30/6000 \times 1000 = 5$, below B’s 18.

B. B recorded 18 new cases per 1000 people in the same 7 days. - Yes: 18 cases among 1000 people, with the period stated.

C. The WHO 75% means 75% of all people died. - The denominator is non-pandemic-related global deaths, not all people.

D. A page published 2025 must contain 2025 measurements. - The source card explicitly describes 2021 data.

10 You do Make a useful infographic

Pupils use paper or a blank document. All sources and data are supplied; an internet connection is optional.

11 You do Your one-page newsroom brief

Supported S1-S4; standard M1-M4; stretch T1-T4. Check the denominator before colour or layout. Accuracy and readability define a useful infographic. Each route supplies chart or response space.

12 Review An editor checks the denominator

Pupils use the audience question to revise an axis label or caption, or justify an accurate one. Quiet review counts. Avoid rewarding dramatic headlines over evidence.

13 Review Publish only what the evidence supports

Model: “B has the highest recorded 7-day cases per 1000 in the constructed example.” Ask pupils to verify the caption retains the constructed-data label.

14 Exit Keep the denominator

E1: 5 per 1000 for the stated period. E2: publication year can differ from data year; data age affects interpretation. E3: no. Re-teach the fraction first if E1 is incorrect.

15 Exit Next useful step

Shared exit minute. Collect the infographic or annotated sketch; keep work private unless the pupil chooses to share.

Answers and assessment

Retrieval

R1: the relevant whole or population. R2: title, category labels, numerical axis with units, period and source. R3: association alone does not prove a cause.

W1-W3

W1: $A = 8$, $B = 18$, $C = 5$ new cases per 1000 people over 7 days. W2: C has the greatest count (30); B has the greatest observed period proportion (18 per 1000). W3: B has the highest recorded proportion; the table has no exposure data or controlled comparison establishing water as its cause.

Hinge

A is false: denominators change the comparison. B is correct: 18 per 1000 for the same 7 days. C is false: WHO uses non-pandemic-related global deaths as the denominator. D is false: publication date and data year can differ.

S1 / M1

A: $24 \div 3000 \times 1000 = 8$. B: $18 \div 1000 \times 1000 = 18$. C: $30 \div 6000 \times 1000 = 5$. These are period proportions per 1000, not person-time rates.

S2 / M2

Correct bars have heights 8, 18 and 5, a zero baseline, town labels A/B/C and units of new cases per 1000 people over 7 days. The headline names B as the highest observed proportion. A hand-drawn chart is valid.

S3 / M3

Example: "B had the highest recorded 7-day cases per 1000 in this constructed example." Source: constructed Town Bulletin. A suitable limit is that the table cannot establish cause; age and exposure may differ, and comparable recording is assumed.

S4 / M4

WHO; publication date 25 September 2025; data year 2021; non-communicable disease deaths; 75% of non-pandemic-related global deaths. This is not 75% of all people. Town Bulletin alternative: teacher-constructed data, three fictional towns, the same 7 days, people at risk at the start. Its calendar publication date is not provided and must not be invented.

T1

D: $6 \div 200 \times 1000 = 30$. E: $30 \div 2000 \times 1000 = 15$. E has the higher count (30 versus 6); D has the higher observed proportion (30 versus 15 per 1000).

T2

$1 \div 200 \times 1000 = 5$ per 1000. One extra recorded case changes D from 30 to 35 per 1000. Comparisons based on small counts can be sensitive to small changes and chance. Uncertainty is not quantified here.

T3

No. Useful missing evidence includes exposure or programme details, a baseline or comparison group, comparable case finding and population characteristics. Other factors may explain a difference. Advanced statistical tests are not required.

T4

Example: “In these constructed 7-day data, D recorded 30 new cases per 1000 and E recorded 15, although E had more cases in total. The comparison describes an observed difference, not why it occurred.”

Exit and assessment

E1: 5 new cases per 1000 over 7 days. E2: data year identifies when the measurements apply. E3: the table supplies no experimental control or causal evidence. L1: accept a specific label, source or caption revision, or a justified confirmation. Secure core requires correct denominators, chart labels and source context. Re-teach by writing $\text{cases} \div \text{population}$ before multiplying by 1000.

Teaching and assessment notes

40 minutes: opening 2, retrieval 3, I do 10, We do 7, hinge 3, independent 11, review 3, exit 1. Zero-minute slides share their stage time.

Choose one response route. Accept accurate speech, pointing plus explanation, annotations or writing as evidence.

A quiet review still has an audience: teacher or TA reads the pupil's decision, asks one evidence question, and the pupil revises or confirms it.

School shorthand may call “cases per 1000 over 7 days” a rate. Strictly this constructed no-baseline-case example is a cumulative incidence proportion for a fixed period; it is not a person-time rate and not an age-standardised comparison.

Interactive We do toggles count and cases per 1000 using exactly A 24/3000, B 18/1000 and C 30/6000. Pupils predict the leading town, test the measure and explain the ranking change. Paper route W1-W3 recreates both comparisons.

The WHO source is used for source literacy, not merged into the fictional infection chart. Do not imply 2021 data describe current school risk.

Access and responsive teaching

Supported

Same core scientific goal, with word bank, short prompts, read-aloud or pointing/dictation. Do not assign a route from diagnosis.

Standard

Use the supplied evidence to explain and justify an independent scientific decision.

Stretch

Optional challenge after secure core: evaluate a limitation and transfer the science. Higher-tier-only material, where present, is labelled explicitly.

Regulation

Preview the sequence. Offer solo or partner work, a pass and quiet response. Allow a brief reset and re-entry at the next numbered task; no compulsory disclosure, performance or speed competition.

Misconceptions to check

The largest case count means the highest population chance.

Counts describe workload; a comparable population denominator is needed for frequency per person.

Check: Compare C's 30/6000 with B's 18/1000.

A publication date is the year the measurements describe.

Data year and publication date can differ.

Check: Ask which year the WHO figure describes.

A difference between towns proves its cause.

It shows a descriptive difference; population makeup, detection and exposures may differ.

Check: Ask what comparison has not been controlled.

75% means 75% of all people had the disease.

The WHO denominator here is non-pandemic-related deaths globally, not all people or all infections.

Check: Require pupils to finish "75% of what?"

Word	Meaning
count	The number of recorded cases or events.
denominator	The population or total used as the bottom of a fraction.
cases per 1000	Cases divided by the relevant population, multiplied by 1000, for a stated period.
data year	The year the data describe; it may differ from publication year.
infographic	A concise visual explanation combining labelled data and words.
association	A pattern in which variables occur together; by itself it does not prove causation.

Scientific references

[Pearson Edexcel GCSE Biology 1 BIO specification](#)

Used to check relevant Foundation curriculum content; this is teacher-authored practice, not an official assessment or complete unit. Checked 8 September 2026.

[WHO: Noncommunicable diseases](#)

Offline source card paraphrases the page dated 25 September 2025: at least 43 million NCD deaths in 2021 and 75% of non-pandemic-related deaths globally. Denominator and data year retained. Checked 8 September 2026.

[CDC Field Epidemiology Manual: Describing Epidemiologic Data](#)

Checked counts versus population comparisons, time/place/person context and chart labelling; lesson arithmetic and fictional town data are original. Checked 8 September 2026.